Imperial College London

Department of Earth Science and Engineering

MSc in Geo-Energy with Machine Learning and Data Science

Independent Research Project

Project Plan

Developing Machine Learning Methods for Crystal Texture Analysis to Investigate Magmatic Evolution and the Onset of Explosive Volcanic Eruptions

by

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GitHub Repository: https://github.com/ese-ada-lovelace-2025/irp-ars223.git

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Abstract

Changes in volcanic crystal textures act as useful records of magmatic processes and their changes across geological time with the potential to offer insights into variations in eruptive behaviour and volcanic hazard assessment. Until now, studying such textures has relied heavily on labour intensive manual petrographic methods which are time consuming and affected by human bias. To expediate and increase the accuracy of such studies this project will develop machine learning methods to automate the analysis of volcanic plagioclase crystal textures. Building upon an existing workflow (PlagNet), plagioclase crystal image data will be extracted from cathodoluminescence (CL) scans via deep-learning-based segmentation models. Self-supervised learning (SSL) will then be applied to the segmented crystal image data in order to learn meaningful crystal-texture representations from the large unlabelled crystal image datasets. Representation learning frameworks such as SimSiam, SimCLR, DINO will be compared for their suitability to this task. Learned embeddings will be analysed with clustering and dimensionality reduction techniques to investigate the correspondence between distinct crystal populations and different magmatic processes. By utilising recent advances in self-supervised learning, this project aims to bypass restrictions created by the need for manually labelled geological datasets to produce previously unreachable insights into textural relationships. The final workflow will be a scalable framework for the quantitative analysis of volcanic crystal populations that will contribute to a broader understanding of the ability of crystal textures to record magmatic evolution.

1. Introduction

1.1 - Problem Outline

Volcanic crystal textures are able to preserve key information regarding magmatic processes and are widely used to understand magma storage conditions, mixing and ascent dynamics in the field of volcanology. Changes in crystal morphology, zoning and population structure are linked to eruptive behaviour. This make them importants indicators for hazard assessment for volcanos (Haller, n.d.; Toth and Maclennan, 2024). Traditional petrographic analysis is a predominantly manual process. This limits the scale at which such analysis can be attempted and the bias in analysis stemming from human observers limits the manual petrographic approach. This is an area of study which because of the reasons outlined above would likely greatly benefit from the automation of such petrographic studies making them more scalable, efficient and accurate.

1.2 - Methodology

1. **Investigating Existing Preprocessing and Segmentation**

CL thin section images will be preprocessed and then segmented to produce plagioclase crystal images using a deep-learning based segmentation model. Post-processing will attempt to remove artefacts and

produce basic morphological descriptors such as crystal size and shape. Although this part of the methodology is already largely complete attempts will be made to optimise and improve the segmentation in order to improve overall results.

2. **Development and Comparison of Self-Supervised Learning (SSL)**

A select range of SSL models (SimSiam, SimCLR, DINO) will be trained the segmented crystal images without manual labels and compared for performance. These models learn the embeddings used in the clustering and analysis stage.

3. **Clustering and Analysis of Embeddings**

The learned embeddings will be analysed with unsupervised clustering methods (i.e. k-means) and dimensionality reduction (i.e. UMAP). This should identify distinct crystal populations and allow the visualisation of structure in the dataset

4. **Optimisation of Chosen Model**

Once a strong model has been identified and initial results have been produced, the model will be optimised in order to refine results.

5. **Interpretation within Geological Context**

Any results suggesting relationships between crystal populations and magmatic processes will be interpreted using knowledge of the data sources and their geological histories.

1.3 - Literature Review

Volcanic crystal textures contain a unique record of pre-eruption magmatic processes and have been used extensively to assess storage conditions, mixing, and eruption dynamics. Changes in crystal textures, such as crystal morphology, zoning and populations, have been correlated with varied eruptive behaviour and analysis of individual crystals may have significant application for understanding volcanic systems and assessing volcanic hazard (Haller, n.d.; Toth and Maclennan, 2024).

More recently, deep learning has been applied to automate the segmentation of crystal phases within cathodoluminescence or thin section imagery. The use of convolutional neural networks has been tested and developed using volcanic and petrographic datasets and has successfully enabled the extraction of individual crystals and the measurement of textural properties (Leichter et al., 2022; Hassan et al., 2024; Halverson et al., 2025; Morales et al., 2025). These studies, while effective and repeatable, usually require large amounts of hand labelled data, the acquisition of which can be prohibitively time consuming and challenging for geological samples.

SSL allows for strong feature representation models to be trained without using any human-labelled data. SSL methods work to exploit the similarities between augmented versions of the same input image (contrastive) or the differences between different images (non-contrastive). Such models include SimCLR (Chen et al., 2020), SimSiam (Chen and He, 2021), BYOL (Grill et al., 2020), DeepCluster (Caron et al., 2018) and SwAV (Caron et al., 2020).

More recent, transformer-based approaches, such as DINO (Caron et al., 2021), DINOv2 (Oquab et al., 2023) and Masked Autoencoders (He et al., 2022) have achieved remarkable success and high levels of generalization across domains.

SSL is becoming increasingly widely used in scientific image analysis where labelled datasets are limited. Uses of SSL models in non-geoscience fields such as histopathology and cell microscopy highlights how they may be used for image data within this project. Studies have shown that self-supervised approaches can be successfully used to reconstruct biologically relevant structure without need for manual annotation (Ciga et al., 2020; Galusser et al., 2023; Bendidi et al., 2024; Moutakanni et al., 2025; Rodgers, 2025). Furthermore, there are some examples of similar studies in adjacent geoscience fields to that of this project like palaeontology and mining geology (Ferreira, 2023; Koeshidayatullah and Ferreira, 2023; Rodgers, 2025). The success demonstrated for other domains suggest the high potential for transferability to geological thin section analysis which faces similar issues regarding data acquisition, complexity, and lack of labels.

Within volcanology, the use of machine learning has expanded significantly in recent years, providing novel ways to analyse volcanic crystals and support the interpretation of volcanic systems. Recent work demonstrates automated crystal population analysis and the measurement of their size distributions for rapid textural investigation (Jutzeler et al., 2024; Toth and Maclennan, 2024; Halverson et al., 2025); however, these methods are mostly supervised

or semi-supervised and hence lack the capacity to efficiently learn from vast unlabelled datasets.

Overall, there is a substantial gap in the literature between the implementation of cutting-edge SSL (Chen et al., 2020; Chen and He, 2021; Caron et al., 2021; Oquab et al., 2023) and its application to volcanic crystal texture analysis (Toth and Maclennan, 2024; Morales et al., 2025). A successful implementation of modern self-supervised representation learning techniques to a segmentation workflow such as PlagNet (Nathwani and Peng, 2026) offers a means to achieve the analysis of crystal textures at an unprecedented scale and hence may provide the capability to explore previously undetectable patterns possibly linked to eruptive behaviour via the clustering of learned embeddings without labelling. This development process will be greatly helped if key lessons learnt from the application of SSL models to similar image data in other fields are transferred to this project where possible (Ciga et al., 2020; Galusser et al., 2023; Ferreira, 2023; Koeshidayatullah and Ferreira, 2023; Bendidi et al., 2024; Moutakanni et al., 2025; Rodgers, 2025).

2. Progress at the Time of This Report

A large dataset of CL thin section images has already been collected and the plagioclase crystal images have been segmented out as part of the work done for the PlagNet repository (Nathwani and Peng, 2026). This means the

foundations of the project have already been developed in the form of the PlagNet repository. PlagNet provides a working pipeline for pre-processing the CL thin-section images and segmenting them. One attempt at producing analysis of the segmented images was attempted using the SimSiam SSL model. Therefore, the project from now will aim to optimise the current segmentation method, improve upon the existing PlagNet pipeline and develop and compare more robust SSL methods such as DINO and SimCLR. PlagNet and all notebooks have been tested to make sure they're in working order and preliminary development of new SSL methods has begun.

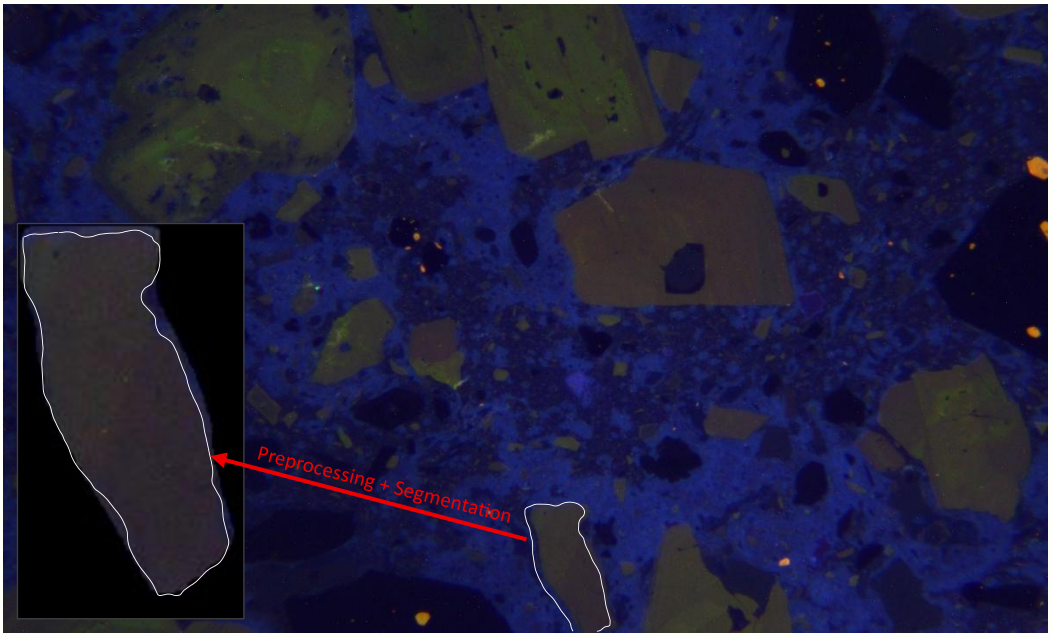


***Figure 1**: Example of raw CL thin section image data (centre) and the segmented plagioclase image data (bottom left) which results from using the pre-exisiting PlagNet pipeline on the raw data.*

3. Plan for the Rest of This Project

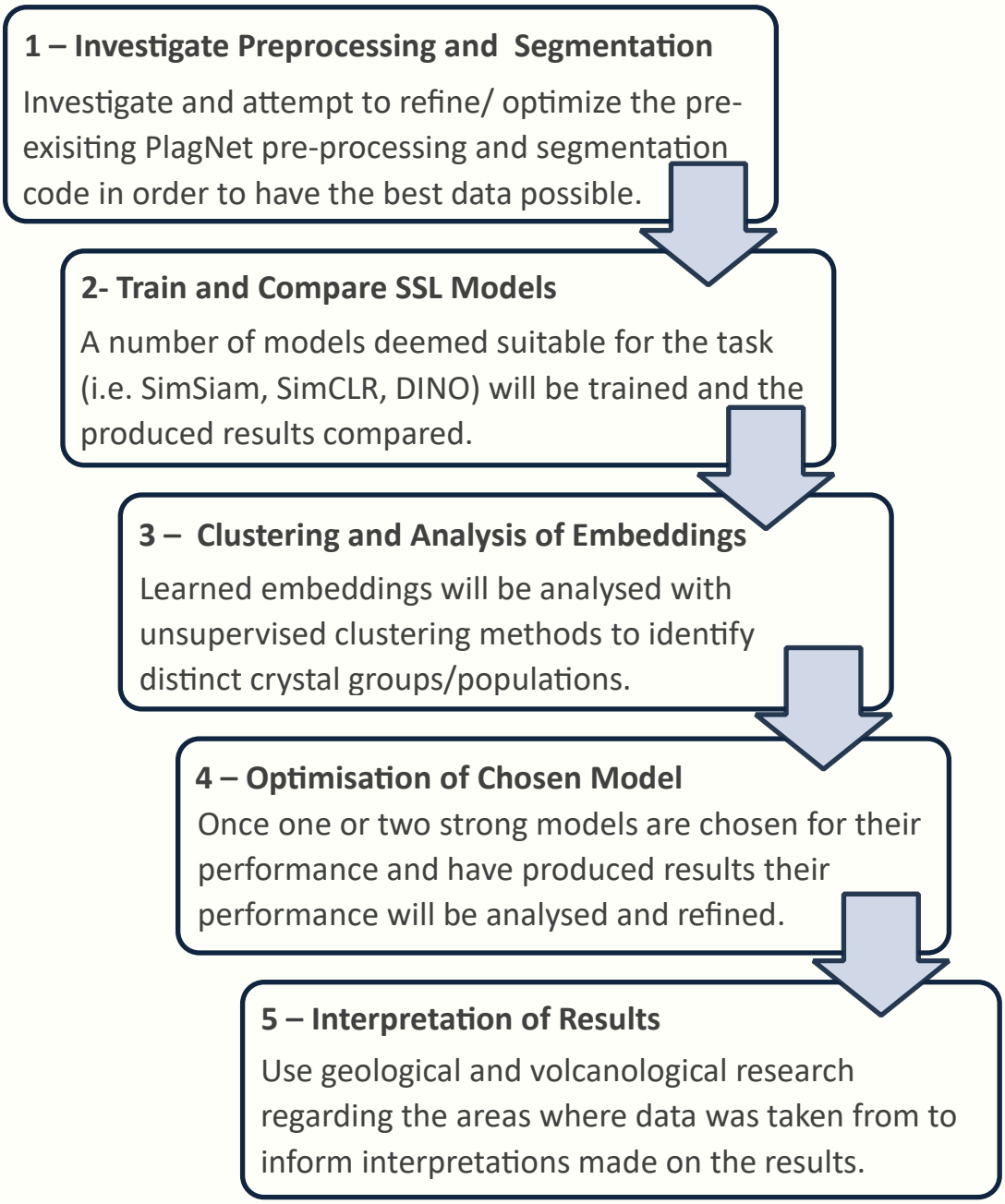


Figure 2: *Flow chart outlining major steps left for the research project.*

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4. AI Acknowledgement

Services: ChatGPT, Claude

Producers: OpenAI, Anthropic

Links: https://chatgpt.com/, https://claude.ai/

How generative AI was used: Used occasionally to understand jargon and notation from some papers which had unfamiliar technical language.

Declaration regarding AI use: All work in this project plan has been produced and written solely by the author listed above despite the use of the Generative AI tool listed above.

5. Notes (DELETE BEFORE FINAL SUBMISSION)

Add figures? – Example of segmentation? Example of SSL model architecture? Some sort of geographical context for Data?

Refine writing? Should be more in-depth or for more general audience?

LIMITATIONS: 1,500 words, 4 Figures max., .pdf file no larger than 4mb